

main.c

Share

Run

Output

Cycle Detected

=== Code Execution Successful

```
1 #include <stdio.h>
2 #define MAX 10
3
4 int adj[MAX][MAX] = {0};
5 int visited[MAX] = {0};
6 void add_edge(int v, int w) {
7
8     adj[v][w] = 1;
9     adj[w][v] = 1;
10 }
11
12 int dfs_cycle(int v, int parent) {
13     visited[v] = 1;
14
15     for (int i = 0; i < MAX; i++) {
16         if (adj[v][i]) {
17             if (!visited[i]) {
18                 if (dfs_cycle(i, v))
19                     return 1;
20             }
21             else if (i != parent) {
22                 return 1;
23             }
24         }
25     }
26     return 0;
27 }
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```
23 }
24 }
25 }
26 return 0;
27 }
28 int detect_cycle(int n) {
29     for (int i = 0; i < n; i++) {
30         if (!visited[i]) {
31             if (dfs_cycle(i, -1))
32                 return 1;
33         }
34     }
35     return 0;
36 }
37 int main() {
38     int n = 3;
39     add_edge(0, 1);
40     add_edge(1, 2);
41     add_edge(2, 0);
42     if (detect_cycle(n))
43         printf("Cycle Detected\n");
44     else
45         printf("No Cycle\n");
46     return 0;
47 }
48 }
```